Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of AI-Powered Sensor Fusion Systems for Real-Time and Reliable Monitoring of Blended Hydrogen			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (✓), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (✓), Cybersecurity ( ), AI (✓), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	The proposed technology is an advanced AI-powered multimodal sensor system designed for the real-time, reliable monitoring of hydrogen–natural gas blended environments in pipelines. • **Multimodal Sensing Architecture:** The system integrates heterogeneous sensing elements—including mass-sensitive, electrochemical, and complementary physical sensors—to capture a multi-dimensional data signature of the gas environment. • **Artificial Intelligence (AI)-Powered Signal Processing:** Unlike traditional sensors, this system utilizes an AI component for sensor fusion, automatic drift compensation, and noise reduction. This allows for high-precision data interpretation even under complex operating conditions such as mixed gas compositions, variable flow regimes, and temperature fluctuations.			

	• **Intelligent Event Classification:** The AI algorithms are trained to distinguish actual leak events from environmental anomalies, significantly reducing false alarms and enabling safer integration of hydrogen into existing natural gas infrastructure.
Rationale for Joint R&D Project	While the University of Calgary research team has successfully demonstrated the core sensing mechanisms and AI-based data interpretation at the laboratory scale (TRL 3-4), international collaboration with a Korean Small and Medium Enterprise (SME) is essential to bridge the gap to commercialization. • **Industrial Engineering & Packaging:** Collaboration is required to transition the delicate laboratory prototype into a rugged, industrial-grade package capable of withstanding real pipeline conditions. • **Field Validation:** The Korean SME partner provides the necessary capabilities for system integration, manufacturability assessment, and validation against industrial safety standards. • **Accelerated Commercialization:** This joint R&D aims to reduce technical and market risks, aligning sensor performance directly with end-user requirements to deliver a practical, commercially ready hydrogen safety solution.
Key Words	1. Hydrogen Blend Monitoring 2. Pipeline Safety Sensors 3. AI-Powered Signal Processing 4. Real-Time Gas Detection 5. Commercial-Ready Sensor Systems

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (7-8)
R&D Objectives	**Primary Objective:** To develop AI-powered multimodal hydrogen sensor systems capable of real-time, accurate, and reliable operation in the hydrogen energy sector. **Specific Objectives:** - **Optimization of Sensing Elements:** Enhance hydrogen sensitivity and selectivity in blended gas environments by optimizing multimodal sensing elements (QCM, IDE, etc.). - **AI Algorithm Development:** Develop AI-powered signal processing algorithms for drift compensation, sensor fusion, and leak-event classification. - **System Integration:** Integrate sensors, data acquisition, and processing modules into deployable monitoring systems. - **Validation:** Validate sensor performance, durability, and reliability under representative operating conditions.			
R&D Target Specifications	The following R&D target specifications are defined in alignment with key Clean Energy Metrics, focusing on hydrogen safety, energy efficiency, and infrastructure reliability for sensor applications. 1) **Hydrogen Detection Range**: 100 ppm – 100 vol%, Broad operational range covering both micro-leaks and full blend ratio monitoring 2) **Response Time**: < 5 seconds, Fast response compatible with emergency shut-off valve (SSV) actuation 3) **Measurement Accuracy**: < ±2% of reading, High-precision measurement for blend management, exceeding standard safety sensors 4) **Selectivity (H_2:CH_4 Ratio)**: > 100:1, High discrimination against methane interference in blended environments via AI fusion 5) **Long-term Stability**: > 6 months maintenance-free, AI-driven drift compensation to extend calibration intervals beyond current industrial norms 6) **Limit of Detection (LOD)**: < 50 ppm, Early warning capability for minor leak events 7) **Operational Conditions**: Temperature: –20 to 80 °C, Humidity: 0 – 95% RH, Pressure: Atmospheric to Pipeline Operating Pressures 8) **System Capability**: Continuous real-time monitoring with automated anomaly detection.			
R&D Plan	This project is structured into four phases to systematically transition the technology from a research prototype to a commercial product. **Phase 1: Sensor and System Optimization** • **Refinement of Multimodal Elements:** Optimize quartz crystal microbalance-based gravimetric sensors, interdigitated electrodes-based electrical sensors, and complementary physical sensors to enhance sensitivity, selectivity, and stability in hydrogen–natural gas blends.			

	• **Surface & Structure Optimization:** Optimize sensor surface coatings and electrode structures to improve response speed, repeatability, and durability under various environmental conditions. • **Cross-Sensitivity Analysis:** Evaluate and mitigate cross-sensitivity effects caused by gas mixtures, odorants, and environmental fluctuations. • **System Architecture Definition:** Define the system architecture for pipeline deployment, covering sensor placement, sampling interfaces, power management, and data acquisition. **Phase 2: AI-Powered Signal Processing Development** • **Dataset Collection:** Systematically collect multimodal sensor datasets under controlled laboratory conditions, covering various gas compositions, flow rates, and environmental factors. • **Sensor Fusion Algorithms:** Develop AI-powered algorithms to integrate heterogeneous sensor outputs into a unified, reliable measurement framework. • **Drift & Noise Compensation:** Implement data-driven models for drift compensation, noise reduction, and baseline correction to ensure stable long-term operation. • **Event Classification:** Develop anomaly detection models to distinguish true hydrogen leak events from environmental disturbances and operational transients. **Phase 3: System Integration and Validation** • **Prototype Assembly:** Assemble integrated sensor systems combining optimized sensing elements, AI modules, and hardware into a unified prototype. • **Operational Integration:** Integrate power management, communication interfaces, and control logic for continuous monitoring. • **Performance Evaluation:** Evaluate performance under simulated pipeline environments, including blended gas exposure and environmental stress conditions. • **Key Performance Index Assessment:** Assess key performance indicators such as sensitivity, response time, accuracy, and stability, followed by iterative optimization. **Phase 4: Commercial Readiness Assessment** • **Manufacturability Review:** Evaluate manufacturability and scalability, including fabrication processes, material availability, and cost considerations. • **Robustness & Compatibility:** Review system robustness, maintenance requirements, and compatibility with existing infrastructure. • **Regulatory & Certification:** Identify technical gaps, regulatory considerations, and certification requirements for field deployment. • **Commercialization Strategy:** Develop pathways for commercialization, including target markets, pilot deployment strategies, and technical documentation for technology transfer.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute will lead the core technology development, scientific validation, and performance benchmarking of the proposed AI-powered multimodal hydrogen sensor systems, building upon validated laboratory-scale and pilot-scale prototypes. **1) Development and optimization of multimodal hydrogen sensing technologies** • Design and refinement of multimodal sensing elements, including mass-sensitive (QCM), electrical (IDE), and complementary sensing modalities, to achieve high sensitivity and selectivity (>100:1) in hydrogen–natural gas blended environments. • Optimization of sensor materials, surface functionalization, and electrode configurations to improve response time (<5s), repeatability, and resistance to environmental interference. • Investigation and mitigation of cross-sensitivity effects arising from gas mixtures, odorants, temperature, and humidity variations. **2) Design and validation of AI-powered signal processing algorithms** • Development of AI-powered algorithms for sensor fusion, enabling robust interpretation of heterogeneous sensor signals. • Implementation of data-driven methods for baseline correction, drift compensation, and noise suppression to support long-term continuous operation (>6 months maintenance-free). • Development and validation of anomaly detection and event classification models to distinguish true hydrogen leak events from operational fluctuations and environmental disturbances. **3) Laboratory and pilot-scale performance evaluation** • Execution of systematic laboratory-scale experiments to characterize sensor performance metrics, including sensitivity, accuracy, response time, repeatability, and durability. • Pilot-scale testing under simulated or representative pipeline operating conditions to assess system robustness and operational stability. **4) Provision of technical expertise and technology transfer** • Development and documentation of standardized testing protocols, calibration procedures, and performance evaluation methodologies. • Preparation of technical documentation, design guidelines, and performance specifications to facilitate effective technology transfer to the SME partner.
Expected R&R of Korean SME Partner	The Korean SME partner will take primary responsibility for industrial system development, validation in real-world operating environments, and commercialization of the proposed hydrogen monitoring technology. **1) Definition of industrial requirements and system specifications** • Identification of end-user requirements based on actual pipeline operation conditions, safety regulations, and industry standards. • Translation of field requirements into system-level performance specifications, including detection thresholds, response time, power consumption, durability, and explosion-proof ratings.

	2) System-level integration, packaging, and engineering design • Integration of sensor modules with industrial-grade data acquisition, power management, and communication interfaces to form deployable sensor systems. • Engineering design of sensor packaging and enclosures suitable for pipeline environments, considering mechanical robustness, environmental protection (IP rating), and installation constraints. **3) Reliability and validation testing under industry-relevant conditions** • Execution of system-level reliability tests, including environmental stress testing, long-term operation, and repeated exposure to hydrogen–natural gas blends. • Validation of system performance under representative operating conditions (Field Trials) to confirm leak detection capabilities and minimize false alarm rates. **4) Contribution to manufacturability and market entry** • Assessment of manufacturability and scalability, including evaluation of fabrication processes, supply chain strategies, and cost structures. • Development of commercialization roadmaps, engagement with potential utility customers, and leadership in pilot demonstrations and product certification.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver significant technological advancements that directly contribute to key Clean Energy Metrics, specifically: • **Unprecedented Leak Detection Speed:** Achieving a response time of less than 5 seconds, enabling immediate actuation of emergency shut-off valves and significantly enhancing pipeline safety compared to conventional sensors. • **High-Precision Blend Monitoring:** Delivering measurement accuracy of < ±2% even in variable flow and blended gas environments, ensuring fair billing and precise process control for hydrogen–natural gas mixtures. • **Robust Selectivity and Reliability:** Securing a hydrogen-to-methane selectivity ratio of >100:1 through AI-driven sensor fusion, effectively eliminating false alarms caused by background natural gas. • **Extended Operational Lifespan:** Demonstrating long-term stability with a maintenance-free operation period of over 6 months via AI-based drift compensation, reducing operational expenditure for utility operators. • **Scalable Commercial Solution:** Delivering a low-power, cost-effective sensor system design ready for mass production and deployment in distributed pipeline monitoring networks. **[Broader Social and Economic Impacts]** From a broader perspective, the project will advance the realization of a hydrogen economy by addressing critical infrastructure challenges:

	• **Strengthening Public Trust & Safety:** By mitigating leak risks and ensuring rapid response to anomalies, the technology will enhance public confidence in the safety of hydrogen infrastructure. • **Enabling Safe Hydrogen Injection:** The technology supports the safe and efficient integration of hydrogen into existing natural gas grids, a crucial step for decarbonizing the heating and energy sectors. • **Global Market Competitiveness:** The collaboration enables the Korean SME partner to secure cutting-edge intellectual properties and validate the technology in global markets, fostering export opportunities in the rapidly growing hydrogen sensor sector. • **Contribution to Net-Zero Goals:** By facilitating the reliable transport and usage of hydrogen, the project directly contributes to emission reduction targets and national clean energy transition strategies.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization